

Module 4 Assignment: Joins and UNIONs
 CPT 166 – Fundamentals of SQL
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1. INNER JOIN – Orders with Customer Names

```
SELECT order_id, order_date, customer_name
FROM orders
INNER JOIN customers ON orders.customer_id = customers.customer_id;
```

	order_id integer	order_date date	customer_name character varying (255)
1	10248	2021-07-04	Wilman Kala
2	10249	2021-07-05	Tradicao Hipermercados
3	10250	2021-07-08	Hanari Carnes
4	10251	2021-07-08	Victuailles en stock
5	10252	2021-07-09	Supremes delices
6	10253	2021-07-10	Hanari Carnes
7	10254	2021-07-11	Chop-suey Chinese
8	10255	2021-07-12	Richter Supermarkt
9	10256	2021-07-15	Wellington Importadora
10	10257	2021-07-16	HILARION-Abastos
11	10258	2021-07-17	Ernst Handel
12	10259	2021-07-18	Centro comercial Moctezuma
13	10260	2021-07-19	Old World Delicatessen
14	10261	2021-07-19	Que Delicia
15	10262	2021-07-22	Rattlesnake Canyon Grocery
16	10263	2021-07-23	Ernst Handel
17	10264	2021-07-24	Folk och fe HB
18	10265	2021-07-25	Blondel pere et fils
19	10266	2021-07-26	Wartian Herkku
20	10267	2021-07-29	Frankenversand
21	10268	2021-07-30	GROSELLA-Restaurante
22	10269	2021-07-31	White Clover Markets
23	10270	2021-08-01	Wartian Herkku
24	10271	2021-08-01	Split Rail Beer & Ale
Total rows: 830		Query complete 00:00:00.055	

2. LEFT JOIN – Customers with No Orders

```
SELECT customer_name, order_id  
FROM customers  
LEFT JOIN orders ON customers.customer_id = orders.customer_id;
```

	customer_name character varying (255) 🔒	order_id integer 🔒
809	Eastern Connection	11056
810	North/South	11057
811	Blauer See Delikatessen	11058
812	Ricardo Adocicados	11059
813	Franchi S.p.A.	11060
814	Great Lakes Food Market	11061
815	Reggiani Caseifici	11062
816	Hungry Owl All-Night Grocers	11063
817	Save-a-lot Markets	11064
818	LILA-Supermercado	11065
819	White Clover Markets	11066
820	Drachenblut Delikatessend	11067
821	Queen Cozinha	11068
822	Tortuga Restaurante	11069
823	Lehmanns Marktstand	11070
824	LILA-Supermercado	11071
825	Ernst Handel	11072
826	Pericles Comidas classicas	11073
827	Simons bistro	11074
828	Richter Supermarkt	11075
829	Bon app	11076
830	Rattlesnake Canyon Grocery	11077
831	Paris specialites	[null]
832	FISSA Fabrica Inter. Salchichas S.A.	[null]
Total rows: 832		Query complete 00:00:00.050

3. RIGHT JOIN – Categories with Products (and those without)

```
SELECT category_name, product_name
FROM products
RIGHT JOIN categories ON products.category_id = categories.category_id
ORDER BY category_name;
```

	category_name character varying (255) 🔒	product_name character varying (255) 🔒
55	Meat/Poultry	Perth Pasties
56	Meat/Poultry	Tourtiaie
57	Meat/Poultry	Alice Mutton
58	Meat/Poultry	Mishi Kobe Niku
59	Meat/Poultry	Pate chinois
60	Meat/Poultry	Thoringer Rostbratwurst
61	Produce	Longlife Tofu
62	Produce	Rassle Sauerkraut
63	Produce	Manjimup Dried Apples
64	Produce	Tofu
65	Produce	Uncle Bobs Organic Dried Pears
66	Seafood	Jacks New England Clam Chow...
67	Seafood	Gravad lax
68	Seafood	Inlagd Sill
69	Seafood	Escargots de Bourgogne
70	Seafood	Ikura
71	Seafood	Nord-Ost Matjeshering
72	Seafood	Carnarvon Tigers
73	Seafood	Konbu
74	Seafood	Red Kaviar
75	Seafood	Boston Crab Meat
76	Seafood	Rogede sild
77	Seafood	Spegesild
78	Unknown	[null]
Total rows: 78		Query complete 00:00:00.056

4. FULL OUTER JOIN – Unmatched Customers and Orders

```
SELECT customer_name, order_id
FROM customers
FULL JOIN orders ON customers.customer_id = orders.customer_id
WHERE customer_name IS NULL
OR order_id IS NULL;
```

	customer_name character varying (255) 🔒	order_id integer 🔒
1	Paris specialites	[null]
2	FISSA Fabrica Inter. Salchichas S.A.	[null]

5. CROSS JOIN – Categories and Test Products

```
SELECT category_name, product_name
FROM categories
CROSS JOIN testproducts
LIMIT 23;
```

	category_name character varying (255) 🔒	product_name character varying (255) 🔒
1	Beverages	Johns Fruit Cake
2	Condiments	Johns Fruit Cake
3	Confections	Johns Fruit Cake
4	Dairy Products	Johns Fruit Cake
5	Grains/Cereals	Johns Fruit Cake
6	Meat/Poultry	Johns Fruit Cake
7	Produce	Johns Fruit Cake
8	Seafood	Johns Fruit Cake
9	Unknown	Johns Fruit Cake
10	Beverages	Janes Favorite Cheese
11	Condiments	Janes Favorite Cheese
12	Confections	Janes Favorite Cheese
13	Dairy Products	Janes Favorite Cheese
14	Grains/Cereals	Janes Favorite Cheese
15	Meat/Poultry	Janes Favorite Cheese
16	Produce	Janes Favorite Cheese
17	Seafood	Janes Favorite Cheese
18	Unknown	Janes Favorite Cheese
19	Beverages	Ellas Special Salmon
20	Condiments	Ellas Special Salmon
21	Confections	Ellas Special Salmon
22	Dairy Products	Ellas Special Salmon
23	Grains/Cereals	Ellas Special Salmon

6. UNION – Unique Product Names

```
SELECT product_name  
FROM products  
UNION  
SELECT product_name  
FROM testproducts  
ORDER BY product_name;
```

	product_name character varying (255) 
1	Alice Mutton
2	Aniseed Syrup
3	Billys Home Made Pizza
4	Boston Crab Meat
5	Camembert Pierrot
6	Carnarvon Tigers
7	Chais
8	Chang
9	Chartreuse verte
10	Chef Antons Cajun Seasoning
11	Chef Antons Gumbo Mix
12	Chocolade
13	Cote de Blaye
14	Elisabeths Best Apples
15	Ellas Special Salmon
16	Escargots de Bourgogne
17	Filo Mix
18	Flotemysost
19	Geitost
20	Genen Shouyu
21	Gnocchi di nonna Alice
22	Gorgonzola Telino
Total rows: 87 Query complete 00	

7. UNION ALL – Duplicates Kept

```
SELECT product_name, 'Products' AS source_table
FROM products
UNION ALL
SELECT product_name, 'Test Products' AS source_table
FROM testproducts
ORDER BY product_name;
```

	product_name character varying (255) 	source_table text 
1	Alice Mutton	Products
2	Aniseed Syrup	Products
3	Billys Home Made Pizza	Test Products
4	Boston Crab Meat	Products
5	Camembert Pierrot	Products
6	Carnarvon Tigers	Products
7	Chais	Products
8	Chang	Products
9	Chartreuse verte	Products
10	Chef Antons Cajun Seasoning	Products
11	Chef Antons Gumbo Mix	Products
12	Chocolade	Products
13	Cote de Blaye	Products
14	Elisabeths Best Apples	Test Products
15	Ellas Special Salmon	Test Products
16	Escargots de Bourgogne	Products
17	Filo Mix	Products
18	Flotemysost	Products
19	Geitost	Products
20	Genen Shouyu	Products
21	Gnocchi di nonna Alice	Products
22	Gorgonzola Telino	Products
Total rows: 87		Query complete 00:00:00.053